Model Card: Review Usefulness Classifier

Built from the ReviewLens experiment on Amazon Electronics reviews. Labels are derived from votes, not human-labelled. Numbers are from the test split only.

1. Model details

Field	Description
Model name	Review usefulness classifier (ReviewLens experiment, unigram config)
Task	Binary: useful (1) vs not useful (0), derived label
Model	TF-IDF (unigrams, min_df=2) + Logistic Regression (C=10)
Training data	Electronics reviews, first 500,000 rows, 17,241 labelled after dedupe (total_votes >= 5)
Test data	70/15/15 stratified, seed 42. Train 12,068, validation 2,586, test 2,587
Labels	helpful_votes / total_votes >= 0.6 (not human-labelled; votes are noisy)
Developed by	ReviewLens project; script `evaluation/one_page_card.py`

2. Intended use

- **Intended:** ranking English product reviews by how useful other shoppers may find them; research and classroom demonstration.
- **Out of scope:** judging individual reviewers; any decision about sellers, products, or customers; other languages; reviews of other categories; fake-review detection.

3. Overall performance

The model reaches 81.8% accuracy and 0.686 macro-F1 on the 2,587 test reviews, against 78.5% accuracy and 0.440 macro-F1 for the majority class baseline.

Model	Accuracy	Class	Precision	Recall	F1
Logistic regression (unigram)	81.8%	not useful (0)	0.618	0.396	0.483
		useful (1)	0.850	0.933	0.890
Majority baseline	78.5%	macro-F1 0.440			

4. Performance across text subgroups

Key audit finding: the model is weakest on reviews that begin with a rating title ("Five Stars ..."). It is also weak on the not-useful class (recall 0.40). The negation subgroup does not show the drop expected from a negation-blind model (see table).

Subgroup	Definition	Test n	Accuracy	Macro-F1
Short reviews	fewer than 20 words	321	67.3%	0.669
Long reviews	20 words or more	2,266	83.9%	0.624
Rating-title reviews	start with 'One..Five Star(s)'	215	69.8%	0.682
Other reviews	no rating title	2,372	82.9%	0.637
Negation words	contains not / never / no / n't	1,664	83.2%	0.668
No negation	none of the above	923	79.3%	0.703

5. Explainability check

Method 1: top coefficients (logistic regression). The strongest words are rare terms such as *intone*, *itunes*, *upon*, *pairs*. They do not describe usefulness. The model fits rare words, so these weights are not evidence of real signal.

Pushes toward useful	Pushes toward not useful
• pictures (+5.21) • upon (+5.13) • original (+4.74) • pairs (+4.74) • previous (+4.61)	• intone (-4.80) • itunes (-4.36) • uncomfortable (-4.31) • not (-4.24) • causing (-4.24)

Method 2: LIME. NOT RUN (lime not installed). No LIME explanation is reported.

6. Limitations and ethical considerations

- **Main limitation:** bag-of-words model. It does not model word order, so "not great" and "great" share features. Use only as a ranking aid.
- **Label noise:** usefulness comes from helpful votes. Votes depend on visibility and popularity, and new reviews have few votes. The 0.6 threshold was chosen by hand, not tuned.
- **Sample:** one category, English only, one week of review dates. The experiment used the first 500,000 rows of the file, not a random sample.
- **Class balance:** the majority baseline reaches 78.5% accuracy, so use macro-F1 for comparison.
- **Data use:** the source data is licensed for academic research only.

7. Recommendations

- Replace the vote-based label with human annotation on a random sample before any claim about usefulness.
- Re-run the subgroup audit on a random sample of the full file and on other categories.
- Try a contextual model (e.g., a fine-tuned DistilBERT) and compare on the same test split and subgroups.